15 There are only n red balls and $(n + 1)$ blue balls in a bag.

Shamsa takes at random 2 balls from the bag.

Show that the probability that both balls are the same colour is $\dfrac{n}{2n+1}$

Total2n+1; after one draw,2n balls remain.

$P(RR) = \dfrac{n}{2n+1} \times \dfrac{n-1}{2n}$

1 mark
Without replacement

$P(BB) = \dfrac{n+1}{2n+1} \times \dfrac{n}{2n}$

$P(same) = \dfrac{n(n-1) + n(n+1)}{2n(2n+1)}$

1 mark
Add RR and BB

$n(n-1) + n(n+1) = 2n^2.$

1 mark
One common fraction

$\dfrac{2n^2}{2n(2n+1)} = \dfrac{n}{2n+1}$

1 mark
Cancel2n;shown

Valid for positive integer n (at least two balls).

(Total for Question 15 is 4 marks)